
Software Design Specifications

for

Food Waste Reducer - Version 1.0

Prepared by:

Gangam Siddanth Reddy(SE23UCSE061): UI/UX and Frontend Engineer

Chalavadi Gayathri(SE23UCSE047): Authentication and User Management Engineer

Duppelli Varshith(SE23UCSE059): Food Inventory and Meal Planner Engineer

Tansihka Guha(SE23UARI124): Notification and Reminder Engineer

Paidy Monisha(SE22UARI118): Local Programs and API Integration Engineer

Katakam Ruthwik(SE23UCSE085): Testing, Quality, Assurance and Deployment Engineer

Gorrepati Aasish Reddy(SE23UCSE070): Database handling Engineer

Goruputi Pranathi(SE23ucse071): Documentation

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Prepared By:

Varshith,Gayathri,Siddanth,Tanishka Guha,Monisha,Aasish, Pranathi,Ruthwik

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1 Introduction

The Software Design Specification (SDS) explains the design of the Food Waste Reducer system. It provides a detailed overview of how the web application functions, what features it has, and how different modules interact with each other. This document is useful for developers, testers and stake holders to understand internal structure of the application .

1.1 Purpose

The purpose of this document is to describe the design and structure of Food Waste Reducer web application. It explains the major system components, interfaces, data base structure and interactions between modules.

1.2 Scope

The Food Waste Reducer system is designed to help user minimize food waste by tracking food inventory, planning meals , receiving expiry reminders, getting food storage tips, and connecting with nearby food - sharing or composting programs.

This System Provides

- User Registration and login
- Food Inventory and Management
- Meal Planning
- Expiry Reminders and Notifications
- Storage Tips For Different Food Items
- Nearby Food Donation and Composting Programs
- Admin Management Features

1.3 Definitions, Acronyms, and Abbreviations

- SDS – Software Design Specification
- UI/UX – User Interface / User Experience
- API – Application Programming Interface
- DB – Database
- Auth – Authentication

1.4 References

- IEEE guidelines for Software Design
- Class notes and materials
- Online sources used for understanding system design

2 Use Case View

This section explains how users will interact with the web application and what actions they can perform. It includes the main features that users will use in the system.

2.1 Use Case

User Registration

- User enters full name, email and password
- System validates details
- User account is created and stored in database

User Login

- User enters email and password
- System checks the credentials
- If correct, user is logged into the application

Add Food Item

- User enters food item details such as name, quantity, category and expiry date
- System stores the food item in inventory

Create Meal Plan

- User selects food items
- User creates a meal plan for a specific date
- System stores the meal plan

Receive Expiry Reminder

- System checks food expiry dates daily
- User receives notifications for items nearing expiry

View Storage Tips

- User selects a food item or food category
- System displays storage tips to extend shelf life

View Nearby Programms

- User allows location access
- System displays nearby food donation centers and composting programs

3 Design Overview

This section provides a high level overview of the Food Waste Reducer system.

The System is divided into multiple modules:

- User Interface Module
- Authentication Module
- Food Inventory Module
- Meal Planner Module
- Notification Module
- Storage Tips Module
- Local Program Finder Module
- Admin Module

These modules interact together to provide the complete functionality of the application.

3.1 Design Goals and Constraints

Design Goals

- Make the system simple and user-friendly
- Help users reduce food waste and effectively
- Provide accurate reminders and notifications
- Offer fast access to local food sharing programs
- Ensure good performance and security

Constraints

- Limited Development Time
- Dependency on third party APIs
- Need to keep the interface simple and lightweight
- Internet connection required for notification and maps

3.2 Design Assumptions

- User have basic knowledge of web applications
- Internet connection is available for notifications and services
- Users enter correct food item information
- Notification services and APIs are available
- The application will run on standard web browser and mobile devices

3.3 Significant Design Packages

The main design packages in the system are:

- User Interface package
- Authentication package

- Food inventory package
- Meal planning package
- Notification package
- Storage tips package
- Local programs package
- Admin package

These modules are connected and work together to form the complete system.

3.4 Dependent External Interfaces

The table below lists the public interfaces this design requires from other modules or applications.

External Application and Interface Name	Module Using the Interface	Functionality/ Description
Firebase cloud messaging(FCM)	Notification Module	Used to send expiry alerts and remainders
Google Maps API	Local program module	Used to find nearby food sharing and composting programs
Weather API	Storage tip module	Used to suggest weather based storage storage recommendations
Authentication service	Login Moudle	Used to verify user credentials
Email service	Authentication module	Used for email verification and password reset

3.5 Implemented Application External Interfaces (and SOA web services)

The table below lists the implementation of public interfaces that this design makes available for other applications.

Interface Name	Module Implementing the Interface	Functionality/ Description
Authentication Interface	Authentication Module	Handles registration, login and profile management
Food inventory	Food inventory module	Handles adding, updating and deleting food items
Meal planner interface	Meal planner Module	Hanldes creating and managing meal plans
Notification interface	Notification module	Sends alerts and remainders

Storage tips interface	Storage tip module	Displays food storage module
Local Program interface	Local program module	Finds nearby food sharing and composting service

4 Logical View

The logical view describes how different layers of application are organized.

- The user interface layer allows user to interact with the application
- The application layer processes user requests and business logic
- The service layer connects APIs, notifications and external services
- The database layer stores all persistent user and food related data

These layers work together to complete the required functions of the system. Each module collaborates with other modules to perform tasks like registration, login, reminders, and emergency alerts. The system design can be further divided into smaller parts if needed, so that each function or method can be clearly explained in a simple way.

4.1 Design Model

- **User Class**

- Stores user details such as name, email and password
- Handles login and profile operations

- **Food Item Class**

- Stores food details such as name, quantity, category and expiry date
- Handles meal scheduling and updates

- **Meal plan Class**

- Stores meal plan details
- Handles meal scheduling and updates

- **Notification Class**

- Stores remainder and notification details
- Handles sending alerts to users

- **Storage Tip Class**

- Stores storage advice for food items
- Display proper food preservation methods

- **Local Program Class**

- Stores nearby donation and composting service details
- Helps user locate nearby programs

- **Admin Class**

- Handles management of users, programs and reports

4.2 Use Case Realization

This section explains how the use cases are implemented in the system.

- For **User Login**:
 - User enters login credentials
 - The authentication module verifies the data
 - Access is granted
- For **Add Food Item**:
 - User enters food item details

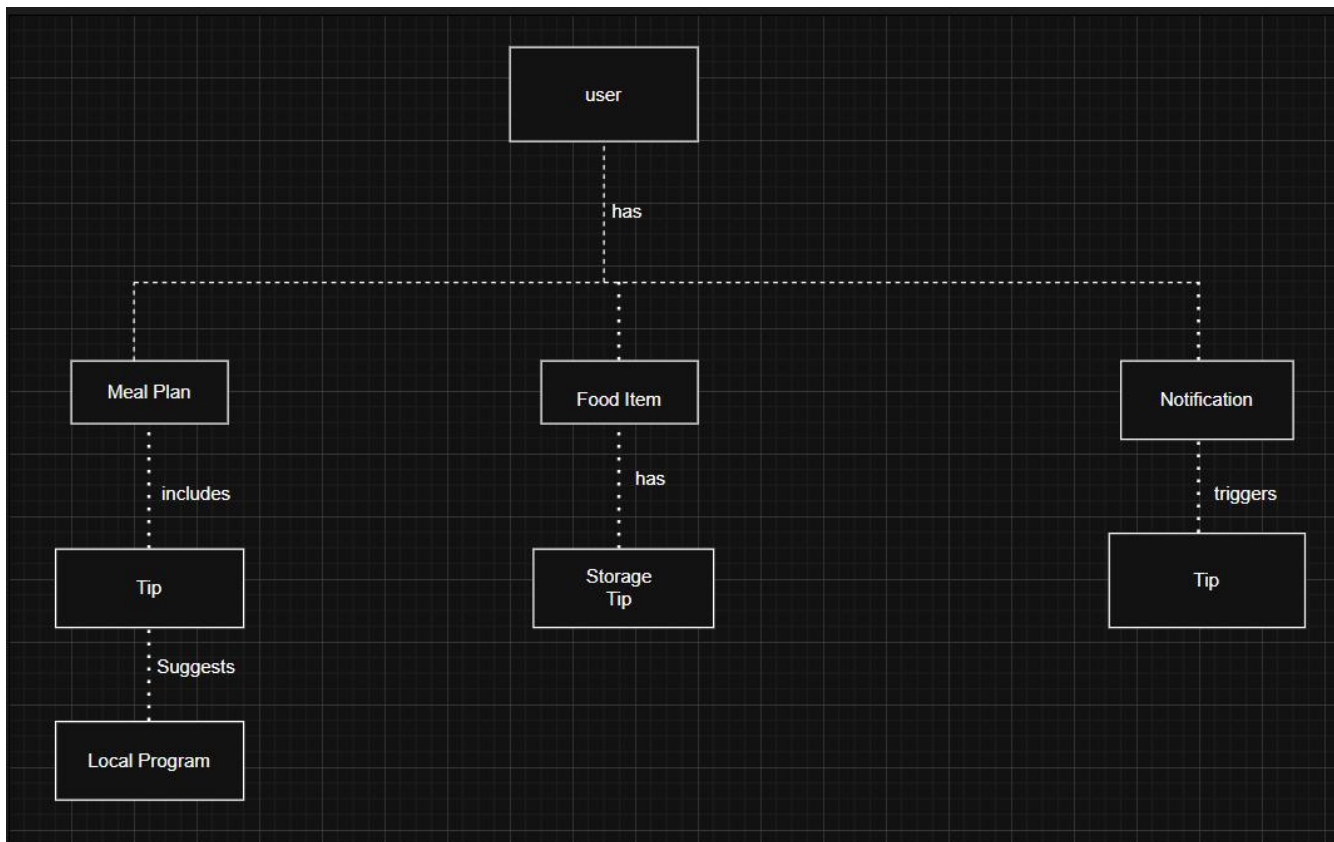
- Food inventory model processes the data
- Data is stored in the database
- For **Expiry Reminder**:
 - System checks food expiry dates
 - Notification module sends alerts to users
- For **Local Program Search**:
 - User location is captured
 - Google maps APIs is used to find nearby programs

At a higher level, modules interact with each other to complete these actions. At a lower level, different classes inside each module work together to perform specific tasks.

5 Data View

This section describes how data is stored and managed in the system. It explains the structure of data and how different data elements are related.

5.1 Domain Model



5.2 Data Model (persistent data view)

The system uses MongoDB collections to share data.

5.2.1 Data Dictionary

- **User Collection** → stores user details
- **Food Item Collection** → stores food inventory details
- **Meal plan collection** → stores meal planning information
- **Notification Collection** → stores alert details
- **Storage Tip Collection** → stores food storage recommendation
- **Local Program Collection** → Stores nearby donation and composting programs

6 Exception Handling

The system handles errors to prevent failures.

- Invalid login credentials show an error message
- Missing food item fields trigger validation messages
- Failed notifications are retried or logged
- Expired or invalid JWT tokens results in access denial
- API failures show appropriate error messages
- Database connection errors are handled gracefully

7 Configurable Parameters

This table describes the simple configurable parameters .

Configuration Parameter Name	Definition and Usage	Dynamic?
Remainder time	User defined time for expiry reminders	Yes

Notification Settings	Controls alerts preferences	yes
Food category filters	Allows users to filter food items	yes
User location	Used to find nearby programs	Yes
Storage tip preferences	Controls type of storage suggestions shown	yes

8 Quality of Service

This section describes how the system maintains availability, security, performance, and monitoring during its usage.

8.1 Availability

The application is designed to be available for users most of the time.

- Supports continuous usage with minimal downtime
- Maintenance updates may require short downtime
- Notification and reminders continue working reliably

8.2 Security and Authorization

The system ensures that user data is secure and protected.

- Users must log in using username and password
- JWT authentication is used for secure access
- Passwords are stored in encrypted form
- Users can only access their own data
- Sensitive data is protected from unauthorized data

8.3 Load and Performance Implications

The system is designed to handle multiple users.

- Can handle simultaneous users efficiently
- Food inventory and notifications load quickly
- Database queries are optimized for performance
- APIs respond quickly to user requests

8.4 Monitoring and Control

The system includes monitoring features to track its performance and usage.

- Errors and failures are logged
- Notifications process are monitored
- API performance is tracked
- Database performance is observed regularly
- Admin panel can be used to manage reports and system activity

